Bundibugyo ebolavirus from the 2026 Ebola outbreak in Uganda and DR Congo: a new variant

Bundibugyo ebolavirus (BDBV) is one of three species members of the genus *Orthoebolavirus* (family *Filoviridae*) that cause disease in humans. The first BDBV outbreak was in the western Uganda district of Bundibugyo (2007–08) and involved 149 cases with 42 fatalities (37 confirmed and five suspected), resulting in a case-fatality rate (CFR) of 36%.[1] The second BDBV outbreak was in Orientale Province, DR Congo, in 2012 and resulted in 38 laboratory-confirmed infections, with 13 laboratory-confirmed fatalities (CFR 34%), and 62 suspected infections, with 34 suspected deaths (CFR 47%).[2] The ongoing outbreak in Ituri Province, DR Congo, has resulted in over 550 suspected infections and 110 deaths, as reported on May 19, 2026.[3] There are currently no approved vaccines, diagnostics, or therapeutics to deploy as medical countermeasures against BDBV. Candidates under early research and development might benefit from refinement to address and tailor them to the observed sequence divergences in circulating 2026 strains.

On May 15, 2026, Uganda declared an imported case from DR Congo of Bundibugyo virus disease with a fatal outcome in a 59-year-old male (CL023114) who was admitted to a private hospital in Kampala on May 11, 2026.[4] The patient deteriorated and died despite intensive care support on May 14, 2026.[5] Based on genome sequencing, sequence analysis, and phylogenetics analysis, we observed that the human isolate was BDBV and is epidemiologically and phylogenetically linked to the ongoing 2026 outbreak in Ituri and Kampala. Our data support that the current outbreak is caused by a new variant of BDBV.

Initial laboratory detection and speciation of the virus identified from the fatal Ugandan case indicated that it was BDBV, based on real-time PCR (rtPCR) performed using the RealStar Filovirus Screen rtPCR Kit 1.0 and RealStar Filovirus Type rtPCR Kit 2.0 (Altona Diagnostics, Hamburg, Germany), together with the BioPerfectus Ebola Virus rtPCR Kit (Jiangsu BioPerfectus Technologies, Taizhou, Jiangsu, China). Assays were performed at the Central Emergency Response and Surveillance Laboratory in Wandegeya, Uganda. Other major viral pathogens were excluded using the Quantiplus Major Fever Panel 1 (Huwel Lifesciences, Hyderabad, India).[6] Using next-generation sequencing on the Illumina NextSeq 2000 Sequencing System (Illumina, San Diego, CA, USA), we obtained a complete genome with 99·7% genome coverage of BDBV (reference genome GenBank accession number NC_014373.1).[6] The sequences from the 2026 BDBV outbreak have since been deposited in the Pathoplexus database (sequence IDs: PP_006XHL9.1 and PP_006XHKB.1 [DR Congo], and PP_006XCJJ.1 [Uganda]) and are publicly accessible.[7]

The consensus genomes of the 2026 BDBV clinical isolates from the DR Congo and Uganda shared between 98·66% and 98·84% identity with the other 17 publicly available BDBV genome sequences of human isolates. Quality control and clade assignment were confirmed using Nextclade (version 3.21.2). At the genome level, the new sequence from Uganda was found to possess a total of 219 single nucleotide polymorphisms when aligned with the reference genome.[7] Examination of the viral open reading frames indicated several amino acid substitutions, with divergences occurring in the genes for glycoprotein, the catalytic subunit of the RNA-dependent RNA polymerase, nucleoprotein, and viral proteins VP24, VP30, VP35, and VP40. There were no major deletions identified; however, a stop codon was noted in the GP_003 at the 367th position.

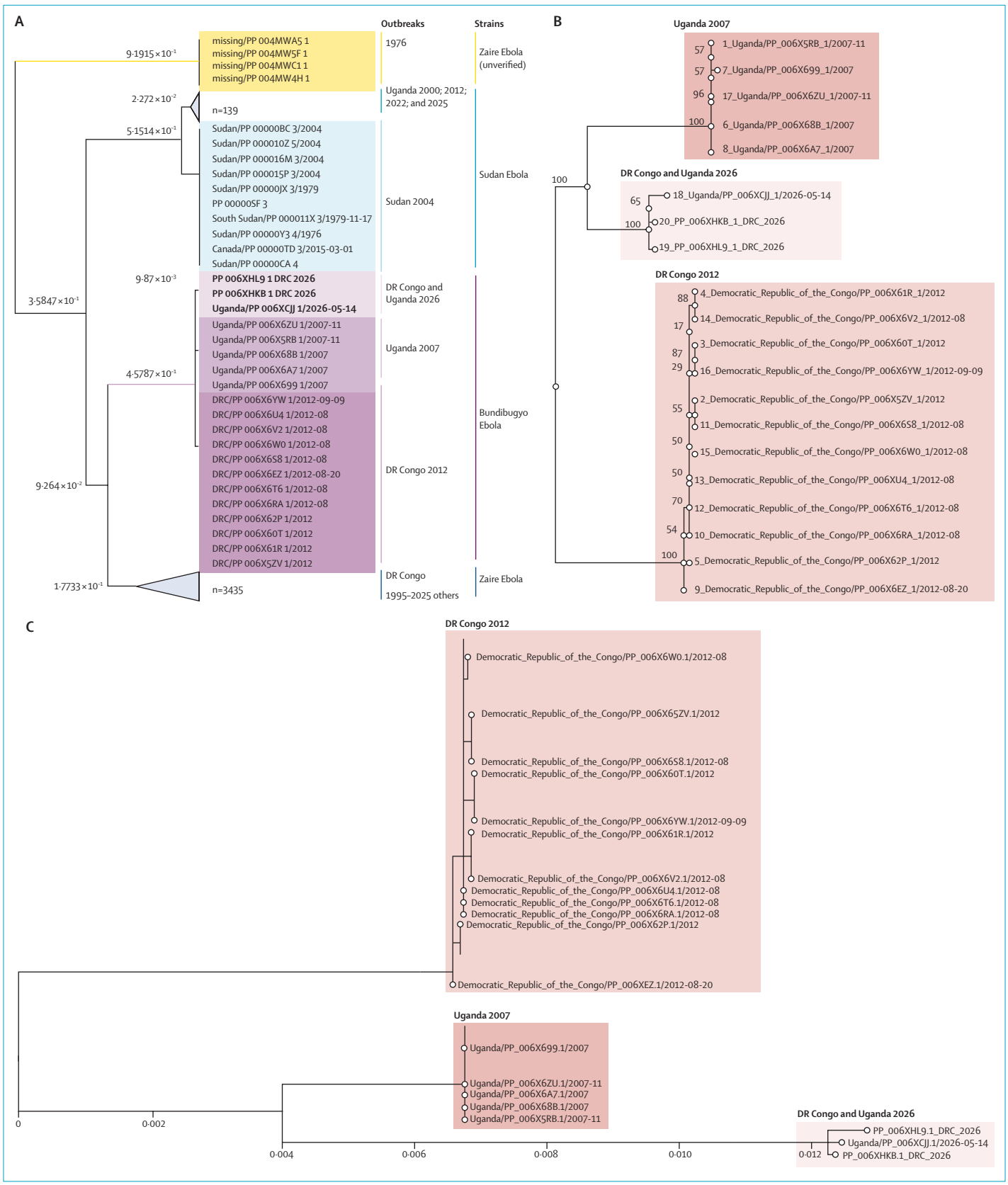
To investigate the phylogeny of the newly sequenced BDBV genome, we constructed a phylogenetic tree using the unweighted pair group method with arithmetic mean (UPGMA) and an alignment score-based approach. As shown in the figure, the newly sequenced Ugandan 2026 BDBV genome clustered with two recently deposited BDBV sequences obtained from patients who were part of the current BDBV outbreak in Ituri. This cluster formed a presumed novel BDBV clade that is phylogenetically distinct from previous outbreaks, including the Kikyo Bundibugyo outbreak (Uganda, 2007–08) and the subsequent Orientale outbreak (DR Congo, 2012; figure A). The 2026 BDBV variant was identified in an individual who had travelled from Ituri to Kampala. It is therefore not surprising that the three 2026 BDBV genomes cluster together, as they are part of the same outbreak. Compared with the old sequences reported from DR Congo in 2012, the new variant showed the closest evolutionary relationship to the sequences from the 2007 Uganda outbreak. To validate the calculated UPGMA tree, two additional phylogenetic trees of the new and previously reported BDBV sequences were also generated using the Nextclade phylogenetic framework (figure B) and neighbour–joining phylogenetic analysis (figure C). These trees also confirmed the distinct evolutionary behaviour of the new BDBV genomes of the ongoing 2026 outbreak, with a divergence rate of 0·0125, which was higher than that of the previous human BDBV isolates. Taken together, the phylogenetic analysis of the 2026 BDBV genomes suggests that the ongoing Bundibugyo virus disease outbreak in DR Congo and Uganda is spreading via a novel BDBV variant.

In conclusion, we show that the BDBV variant identified in Uganda

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A
Outbreaks
Strains
missing/PP 004MWA5 1
missing/PP 004MW5F 1
missing/PP 004MWC1 1
missing/PP 004MW4H 1
1976
Zaire Ebola (unverified)
9·1915×10⁻¹
2·272×10⁻²
n=139
Uganda 2000; 2012; 2022; and 2025
5·1514×10⁻¹
Sudan/PP 00000BC 3/2004
Sudan/PP 000010Z 5/2004
Sudan/PP 000016M 3/2004
Sudan/PP 000015P 3/2004
Sudan/PP 00000JX 3/1979
PP 00000SF 3
South Sudan/PP 000011X 3/1979-11-17
Sudan/PP 00000Y3 4/1976
Canada/PP 00000TD 3/2015-03-01
Sudan/PP 00000CA 4
Sudan 2004
Sudan Ebola
9·87×10⁻³
PP 006XHL9 1 DRC 2026
PP 006XHKB 1 DRC 2026
Uganda/PP 006XCJJ 1/2026-05-14
DR Congo and Uganda 2026
3·5847×10⁻¹
Uganda/PP 006X6ZU 1/2007-11
Uganda/PP 006X5RB 1/2007-11
Uganda/PP 006X68B 1/2007
Uganda/PP 006X6A7 1/2007
Uganda/PP 006X699 1/2007
Uganda 2007
4·5787×10⁻¹
DRC/PP 006X6YW 1/2012-09-09
DRC/PP 006X6U4 1/2012-08
DRC/PP 006X6V2 1/2012-08
DRC/PP 006X6W0 1/2012-08
DRC/PP 006X6S8 1/2012-08
DRC/PP 006X6EZ 1/2012-08-20
DRC/PP 006X6T6 1/2012-08
DRC/PP 006X6RA 1/2012-08
DRC/PP 006X62P 1/2012
DRC/PP 006X60T 1/2012
DRC/PP 006X61R 1/2012
DRC/PP 006X5ZV 1/2012
DR Congo 2012
Bundibugyo Ebola
9·264×10⁻²
1·7733×10⁻¹
n=3435
DR Congo
1995–2025 others
Zaire Ebola
B
Uganda 2007
57 1_Uganda/PP_006X5RB_1/2007-11
57 7_Uganda/PP_006X699_1/2007
96 17_Uganda/PP_006X6ZU_1/2007-11
100 6_Uganda/PP_006X68B_1/2007
8_Uganda/PP_006X6A7_1/2007
100
DR Congo and Uganda 2026
65 18_Uganda/PP_006XCJJ_1/2026-05-14
100 20_PP_006XHKB_1_DRC_2026
19_PP_006XHL9_1_DRC_2026
DR Congo 2012
88 4_Democratic_Republic_of_the_Congo/PP_006X61R_1/2012
17 14_Democratic_Republic_of_the_Congo/PP_006X6V2_1/2012-08
87 3_Democratic_Republic_of_the_Congo/PP_006X60T_1/2012
29 16_Democratic_Republic_of_the_Congo/PP_006X6YW_1/2012-09-09
2_Democratic_Republic_of_the_Congo/PP_006X5ZV_1/2012
55 11_Democratic_Republic_of_the_Congo/PP_006X6S8_1/2012-08
50 15_Democratic_Republic_of_the_Congo/PP_006X6W0_1/2012-08
50 13_Democratic_Republic_of_the_Congo/PP_006XU4_1/2012-08
70 12_Democratic_Republic_of_the_Congo/PP_006X6T6_1/2012-08
54 10_Democratic_Republic_of_the_Congo/PP_006X6RA_1/2012-08
100 5_Democratic_Republic_of_the_Congo/PP_006X62P_1/2012
9_Democratic_Republic_of_the_Congo/PP_006X6EZ_1/2012-08-20
C
DR Congo 2012
Democratic_Republic_of_the_Congo/PP_006X6W0.1/2012-08
Democratic_Republic_of_the_Congo/PP_006X65ZV.1/2012
Democratic_Republic_of_the_Congo/PP_006X6S8.1/2012-08
Democratic_Republic_of_the_Congo/PP_006X60T.1/2012
Democratic_Republic_of_the_Congo/PP_006X6YW.1/2012-09-09
Democratic_Republic_of_the_Congo/PP_006X61R.1/2012
Democratic_Republic_of_the_Congo/PP_006X6V2.1/2012-08
Democratic_Republic_of_the_Congo/PP_006X6U4.1/2012-08
Democratic_Republic_of_the_Congo/PP_006X6T6.1/2012-08
Democratic_Republic_of_the_Congo/PP_006X6RA.1/2012-08
Democratic_Republic_of_the_Congo/PP_006X62P.1/2012
Democratic_Republic_of_the_Congo/PP_006XEZ.1/2012-08-20
Uganda 2007
Uganda/PP_006X699.1/2007
Uganda/PP_006X6ZU.1/2007-11
Uganda/PP_006X6A7.1/2007
Uganda/PP_006X68B.1/2007
Uganda/PP_006X5RB.1/2007-11
DR Congo and Uganda 2026
PP_006XHL9.1_DRC_2026
Uganda/PP_006XCJJ.1/2026-05-14
PP_006XHKB.1_DRC_2026
0
0·002
0·004
0·006
0·008
0·010
0·012

from a travel-related case linked to the ongoing outbreak in Ituri represents a new clade of BDBV. These findings underscore the diversity of the BDBV ecosystem and the urgent need for antiviral and vaccine countermeasures to reduce fatalities and for public health measures to limit the spread of the virus. Global efforts are immediately needed in this region. Continued genomic surveillance is required for in-depth monitoring to track emerging BDBV variants and adjust and employ BDBV countermeasures in a timely manner.

MW, AK, HKB, WS, and DJK conceptualised the study. AA, AN, IS, SN, SK, VNZ, and MW sequenced the Uganda BDBV genome. AA-A, TW-B, EK-L, DM-B, NM-M, GL-N, and PM-K sequenced and performed the bioinformatics analysis of BDBV genomes from DR Congo. AK and MW analysed the genomics data and phylogeny. AN, SN, HKB, WS, BB, BK, MD, GSM, CO, J-ROA, and DA analysed the data. AN, AK, HKB, WS, DJK, WM, and MW wrote the original version of the Correspondence. TW-B, AK, DJK, and MW revised the Correspondence. All authors equally contributed to reviewing and editing processes, and approved the final version of the Correspondence. AA and AA-A are joint first authors; PM-K and MW jointly supervised the study. All authors had full access to all the data in the study and had final responsibility for the decision to submit for publication.

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Figure: **Phylogenetic tree of the relationship of circulating BDBV to previous BDBV isolates**
Relationships between the 2026 BDBV variant, responsible for the 2026 Bundibugyo virus disease outbreak in DR Congo and Uganda, and previous BDBV variants and other ebolavirus species of public interest in the region (Zaire ebolavirus [*Orthoebolavirus zairense*] and Sudan ebolavirus [*Orthoebolavirus sudanense*]). (A) Unweighted pair group method with arithmetic mean phylogenetic tree of the sequences from the ongoing BDBV outbreak and other strains from the previous outbreaks. (B) Calculated neighbour–joining phylogenetic tree of the BDBV sequences representing the new subclade from the ongoing outbreak and two BDBVs from the previous outbreaks. The neighbour–joining tree was calculated using the Jukes–Cantor model with 1000 bootstrap resampling. (C) Representation of the calculated phylogenetic tree of the BDBV sequences from the ongoing outbreak and previous outbreaks highlighting the divergence score. BDBV=Bundibugyo ebolavirus.

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Pregnancy as exclusion criterion in Ebola research: not again

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As the 2026 Bundibugyo virus (*Orthoebolavirus bundibugyoense*) Public Health Emergency of International Concern progresses,[1] we draw attention to key lessons from the 2014–16 Ebola (*Orthoebolavirus zairense*) outbreak in Guinea, Liberia, and Sierra Leone.[2] One crucial learning was that women and gender-diverse people who were pregnant or breastfeeding were systematically excluded from most interventional trials.[2] Exclusion has historically been the norm for protecting fetuses and those who are pregnant,[3] however, if applied without contextualised justification, exclusion can harm instead of protect. During the 2014–16 Ebola virus outbreak, pregnancy was correlated with the highest mortality rates, and fetal mortality was 100% when